Cleaning Substation Data for Reliable Anomaly Detection

Fixing missing values, duplicates, and time irregularities

Sense Reply × Albert School – BDD Challenge

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Understanding the project's foundation and goals

Context

Context

Substation consumption data was collected through SCADA, IoT Gateway, and Cloud, revealing transmission noise and structural issues that required addressing for accurate analysis.

Mission

The primary goal was to clean and preprocess the dataset to enable reliable anomaly detection, ensuring that the data could be trusted for further analysis and insights.

Final Output

The result was a fully cleaned dataset with consistent 10-minute frequency, free from gaps and duplicates, while also incorporating quality assurance features to enhance reliability.

Datetime	PowerConsumption_Zone	PowerConsum
2017-01-01 00:00:00	34055.6962	1
2017-01-01 00:10:00	29814.68354	1
2017-01-01 00:20:00	29128.10127	1
2017-01-01 00:30:00	28228.86076	1
2017-01-01 00:40:00	27335.6962	1
2017-01-01 00:50:00	26624.81013	1
2017-01-01 01:00:00	25998.98734	1
2017-01-01 01:10:00	25446.07595	1
2017-01-01 01:20:00	24777.72152	1
2017-01-01 01:30:00	24279.49367	1
2017-01-01 01:40:00	23896.70886	1
2017-01-01 01:50:00	23544.3038	1
2017-01-01 02:00:00	23003.5443	1
2017-01-01 02:10:00	22329.11392	
2017-01-01 02:20:00	22092.1519	
2017-01-01 02:30:00	21903.79747	1
2017-01-01 02:40:00	21685.06329	1
2017-01-01 02:50:00	21484.55696	1
2017-01-01 03:00:00	21107.8481	1
2017-01-01 03:10:00	20989.36709	1
2017-01-01 03:20:00	20870.88608	1
2017-01-01 03:30:00	20870.88608	1
2017-01-01 03:40:00	20597.46835	1
2017-01-01 03:50:00	20421.26582	1
2017-01-01 04:00:00	20524.55696	1
2017-01-01 04:10:00	20[illegible]	1
2017-01-01 04:20:00	205[illegible]	1

Problem 1: Missing Values: 4320

```
substation_data[power_columns] = substation_data[power_columns].interpolate(method="time")
```

Dataset Problem

Missing timestamps and consumption values lead to breaks in time-series continuity, making anomaly detection unreliable.

Dataset Solution

Reindexing the full time range and filling gaps through interpolation ensures a complete dataset for analysis.

Physical Cause

SCADA polling delays, IoT Gateway packet loss, and device communication drops contribute to missing data issues.

Preventive Measures

Implementing heartbeat signals and redundant buffering can help mitigate missing data and enhance reliability.

Problem 2: Duplicates: 1493

```
rows_before = len(substation_data)
substation_data = substation_data[~substation_data.index.duplicated()]
rows_after = len(substation_data)
print(f"Removed {rows_before - rows_after} duplicate rows.\n")
```

Dataset Problem

Multiple identical timestamps in the raw dataset create **inconsistent time-series behavior**, undermining the integrity of the analysis.

Dataset Solution

To address these issues, remove duplicate rows to preserve **one reading per timestamp**, ensuring reliability in data interpretation.

Physical Cause

The presence of duplicates often arises from retransmission of packets due to **communication retries** or device logging loops.

Preventive Measures

Implementing sequence numbers on each data point helps prevent duplicates, alongside **deduplication logic** at the Edge during data ingestion.

Problem 3: Irregular Time Intervals

```
substation_data = substation_data.resample('10min').mean()
substation_data.info()
substation_data
```

Dataset Problem

Data did not adhere to the expected sampling rate, causing significant **gaps** in time-series continuity.

Dataset Solution

Implementing **resampling** techniques helped to enforce a consistent 10-minute frequency for the dataset.

Physical Cause

Timing issues stemmed from SCADA irregularities and **incorrect device clocks**, leading to erratic data arrival.

Preventive Measures

To mitigate future discrepancies, **NTP time synchronization** and consistent polling checks were established.

Additional Checks

Enhancements for Data Quality

```
substation_data["Hour"] = substation_data.index.hour
substation_data["DayOfWeek"] = substation_data.index.dayofweek #0 = Monday
substation_data["IsWeekend"] = (substation_data["DayOfWeek"] >= 5).astype(int)
```

New Columns

Additional columns were introduced, such as hour, day, weekday, and weekend flags. These features are instrumental in enabling better insights into patterns and usage.

Rolling Statistics

A rolling standard deviation was calculated to monitor fluctuations over time. This process is crucial for identifying any significant deviations.

Anomaly Detection

Anomaly flags, both strict and relaxed, were added to the dataset. These flags enhance our ability to detect unusual behavior and improve overall model performance.

```
TotalConsumption,Hour,DayOfWeek,IsWeekend,Total_rolling_1h,Anomaly_flag,anomaly_relaxed
```

CONCLUSION

Summary of Key Achievements

Clean Dataset

We achieved a **clean, structured dataset** with a consistent 10-minute time series, ensuring there are no missing values, duplicates, or irregular intervals, enhancing data reliability.

Anomaly Detection

The dataset is now **fully prepared** for anomaly detection, with additional quality-control features and flags implemented, allowing for better operational decisions based on accurate insights.